

Assignment II

LUKE PROFIO

Query 1 – Insurance Distribution Among ICU Admissions

- Provides an overview of ICU admissions by insurance type.
- Uses GROUP BY on the insurance field to count admissions per insurer. This includes a subquery which computes the percentage of total admissions for each insurance category.
- Roughly half of ICU patients are covered by Medicare, followed by private insurance. Medicare and private insurance cover almost 90% of ICU patients.

Query 1 - Results

Row	insurance	num_admissions	perc_of_admissions
1	Medicare	28215	47.8
2	Private	22582	38.3
3	Medicaid	5785	9.8
4	Government	1783	3.0
5	Self Pay	611	1.0

Query 1

```
SELECT insurance, COUNT(*) AS num_admissions,
ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM `physionet-
  data.mimiciii_clinical.admissions`), 1) AS perc_of_admissions
FROM `physionet-data.mimiciii_clinical.admissions`
GROUP BY insurance
ORDER BY num_admissions DESC;
```

Query 2 – Admission Sources by Insurance Type

- This query examines how patients with different insurances enter the hospital (e.g. emergency department, transfer), to see if the payer type correlates with the admission source.
- The query groups admissions by both insurance and admission_location, along with the entries for each combination. Results are ordered to list the most common admission sources for each insurance category.
- In all but private insurance, patients primarily enter the hospital through emergency room admissions.

Query 2 – Results

Row	insurance	admission_location	admission_count
1	Government	EMERGENCY ROOM ADMIT	672
2	Government	PHYS REFERRAL/NORMAL DELI	469
3	Government	CLINIC REFERRAL/PREMATURE	428
4	Government	TRANSFER FROM HOSP/EXTRA...	198
5	Government	** INFO NOT AVAILABLE **	8
6	Government	HMO REFERRAL/SICK	5
7	Government	TRANSFER FROM SKILLED NUR	2
8	Government	TRANSFER FROM OTHER HEALT	1
9	Medicaid	EMERGENCY ROOM ADMIT	2357
10	Medicaid	PHYS REFERRAL/NORMAL DELI	1417
11	Medicaid	CLINIC REFERRAL/PREMATURE	1343
12	Medicaid	TRANSFER FROM HOSP/EXTRA...	580
13	Medicaid	** INFO NOT AVAILABLE **	43
14	Medicaid	HMO REFERRAL/SICK	27
15	Medicaid	TRANSFER FROM SKILLED NUR	11
16	Medicaid	TRANSFER FROM OTHER HEALT	7
17	Medicare	EMERGENCY ROOM ADMIT	12808
18	Medicare	CLINIC REFERRAL/PREMATURE	5653

Query 2 -

```
SELECT insurance, admission_location, COUNT(*) AS admission_count
FROM `physionet-data.mimiciii_clinical.admissions`
GROUP BY insurance, admission_location
ORDER BY insurance, admission_count DESC;
```

Query 3 – Discharge Disposition by Insurance Type

- This query shows where patients go after their hospital/ICU stay, and how this varies by insurance type.
- We use a GROUP BY on insurance and discharge_location to count the total number of discharges. Then we sort by count to highlight the most common discharge outcomes per insurance group.
- Across all insurance types, patients are predominantly discharged to their home.

Query 3 – Results

Row	insurance	discharge_location	patient_count
1	Government	HOME	1030
2	Government	HOME HEALTH CARE	338
3	Government	DEAD/EXPIRED	90
4	Government	REHAB/DISTINCT PART HOSP	89
5	Government	SHORT TERM HOSPITAL	63
6	Government	DISCH-TRAN TO PSYCH HOSP	45
7	Government	SNF	39
8	Government	LONG TERM CARE HOSPITAL	27
9	Government	DISC-TRAN CANCER/CHLDRN H	23
10	Government	LEFT AGAINST MEDICAL ADVI	23
11	Government	HOSPICE-HOME	10
12	Government	OTHER FACILITY	3
13	Government	ICF	1
14	Government	HOME WITH HOME IV PROVIDR	1
15	Government	HOSPICE-MEDICAL FACILITY	1
16	Medicaid	HOME	2509
17	Medicaid	HOME HEALTH CARE	1232
18	Medicaid	REHAB/DISTINCT PART HOSP	389
19	Medicaid	DEAD/EXPIRED	381
20	Medicaid	SNF	335
21	Medicaid	SHORT TERM HOSPITAL	284

Query 3 -

```
SELECT insurance, discharge_location, COUNT(*) AS patient_count
FROM `physionet-data.mimiciii_clinical.admissions`
GROUP BY insurance, discharge_location
ORDER BY insurance, patient_count DESC;
```

Query 4 – ICU Transfers by Insurance Type

- Here we measure ICU utilization intensity by insurance type, especially to find how often patients had multiple ICU stays during a single hospital admission.
- In the query we JOIN admissions with ICU stays on hospital admission ID, so each ICU stay is linked to the patient's insurance. Then we use the COUNT and DISTINCT counts to calculate the average number of ICU stays per admission for each insurance category.
- Medicare has the highest average number of ICU stays per admission, followed by Medicaid.

Query 4 – Results

Row	insurance	total_icu_stays	total_admissions	avg_icu_stays_per_admission
1	Medicare	29812	27642	1.08
2	Private	23244	22097	1.05
3	Government	1837	1758	1.04
4	Medicaid	6017	5683	1.06
5	Self Pay	622	606	1.03

Query 4 -

```
SELECT A.insurance,
COUNT(I.icustay_id) AS total_icu_stays,
COUNT(DISTINCT A.hadm_id) AS total_admissions,
ROUND(COUNT(I.icustay_id) * 1.0 / COUNT(DISTINCT A.hadm_id), 2) AS
  avg_icu_stays_per_admission
FROM `physionet-data.mimiciii_clinical.admissions` A
JOIN `physionet-data.mimiciii_clinical.icustays` I
ON A.hadm_id = I.hadm_id
GROUP BY A.insurance;
```

Query 5 – Average ICU Length of Stay by Insurance Type

- This query compares the average ICU length of stay (LOS) for patients by insurance type.
- A subquery first calculates each admission's total ICU LOS by summing los across all ICU stays in that hospitalization . The outer query then groups by insurance and averages those totals. This makes sure multi-ICU stays are combined per admission before averaging.
- Medicaid has the highest average ICU length of stay, followed by private insurance.

Query 5 – Results

Row	insurance	avg_icu_days
1	Private	5.9
2	Medicare	4.56
3	Medicaid	6.21
4	Government	5.11
5	Self Pay	3.13

Query 5 -

```
SELECT insurance, ROUND(AVG(total_icu_days), 2) AS avg_icu_days
FROM (
    SELECT A.hadm_id, A.insurance, SUM(I.los) AS total_icu_days
    FROM `physionet-data.mimiciii_clinical.admissions` A
    JOIN `physionet-data.mimiciii_clinical.icustays` I
        ON A.hadm_id = I.hadm_id
    GROUP BY A.hadm_id, A.insurance
) AS per_adm
GROUP BY insurance;
```

Query 6 – Average Number of Procedures by Insurance Type

- Here we evaluate how many procedures patients undergo on average per admission for each insurance category. This reveals the utilization intensity by payer group.
- A subquery counts procedures for each hospital admission. We LEFT JOIN this to the admissions table to include admissions with zero procedures. Then we group by insurance and take the average count of procedures per admission.
- Medicare has the highest average number of procedures, followed by private insurance.

Query 6 – Results

Row	insurance	avg_procedures_per_admission
1	Medicare	4.27
2	Private	3.98
3	Government	3.7
4	Medicaid	3.68
5	Self Pay	3.35

Query 6 -

```
SELECT A.insurance,
      ROUND(AVG(COALESCE(proc.num_procedures, 0)), 2) AS avg_procedures_per_admission
FROM `physionet-data.mimiciii_clinical.admissions` A
LEFT JOIN (
    SELECT hadm_id, COUNT(*) AS num_procedures
    FROM `physionet-data.mimiciii_clinical.procedures_icd`
    GROUP BY hadm_id
) AS proc
ON A.hadm_id = proc.hadm_id
GROUP BY A.insurance;
```

Query 7 – Most Common Procedure for Each Insurance Category

- Here we identify the most frequently performed procedure (by ICD-9 code) for each insurance group, showing the most common treatments/procedures for each patient segment.
- We use a common table expression, proc_counts, to count occurrences of each procedure code per insurance. A second CTE max_proc finds the maximum count per insurance. The overall query joins these to show the procedure code(s) with the max count for each insurance.
- Venous catheterization is the most common procedure for Medicare (highest count) and Medicaid.

Query 7 – Results

Row	insurance	icd9_code	top_procedure_count
1	Private	9955	4538
2	Medicare	3893	8465
3	Medicaid	3893	1516
4	Government	9671	326
5	Self Pay	9671	215

Query 7 -

```
WITH proc_counts AS (
  SELECT A.insurance, P.icd9_code, COUNT(*) AS proc_count
  FROM `physionet-data.mimiciii_clinical.admissions` A
  JOIN `physionet-data.mimiciii_clinical.procedures_icd` P
      ON A.hadm_id = P.hadm_id
  GROUP BY A.insurance, P.icd9_code
), max_proc AS (
  SELECT insurance, MAX(proc_count) AS max_count
  FROM proc_counts
  GROUP BY insurance
)
SELECT pc.insurance, pc.icd9_code, pc.proc_count AS top_procedure_count
FROM proc_counts pc
JOIN max_proc mp
    ON pc.insurance = mp.insurance AND pc.proc_count = mp.max_count;
```

Query 8 – Average Hospital Length of Stay by Insurance Type

- Here we compare the average total hospital length of stay for patients by insurance type.
- We calculate the duration in days between admission and discharge (TIMESTAMP_DIFF) for each record, then use GROUP BY insurance with an AVG to find the mean stay length for each insurance category.
- Medicaid holds the highest average total hospital length of stay, followed by private insurance.

Query 8 – Results

Row	insurance	avg_hospital_LOS_days
1	Medicare	9.88
2	Private	10.14
3	Government	9.73
4	Medicaid	11.44
5	Self Pay	6.34

Query 8 -

```
SELECT insurance,
    ROUND(AVG(TIMESTAMP_DIFF(dischtime, admittime, DAY)), 2) AS avg_hospital_LOS_days
FROM `physionet-data.mimiciii_clinical.admissions`
GROUP BY insurance;
```

Query 9 – In-Hospital Mortality Rate by Insurance Type

- Here we calculate the in-hospital mortality for each insurance group – the percentage of admissions where the patient died before discharge. This indicates if mortality varies with insurance.
- We use the hospital_expire_flag in a SUM to count deaths per insurance, divided by total admissions per insurance. The GROUP BY shows mortality counts and computes a percentage.
- Medicare and self-pay cohorts hold the highest in-hospital mortality rate.

Query 9 – Results

Row	insurance	deaths	total_admissions	mortality_rate_pct
1	Medicare	3905	28215	13.84
2	Private	1383	22582	6.12
3	Government	90	1783	5.05
4	Medicaid	381	5785	6.59
5	Self Pay	95	611	15.55

Query 9 -

```
SELECT insurance,
    SUM(CASE WHEN hospital_expire_flag = 1 THEN 1 ELSE 0 END) AS deaths,
    COUNT(*) AS total_admissions,
    ROUND(100 * SUM(CASE WHEN hospital_expire_flag = 1 THEN 1 ELSE 0 END) / COUNT(*),
2) AS mortality_rate_pct
FROM `physionet-data.mimiciii_clinical.admissions`
GROUP BY insurance;
```

Query 10 – 30-Day Readmission Rate by Insurance Type

- Here we pull data from the admissions table to find the 30-day readmission rate by insurance type.
- For each hospital stay, we use a nested EXISTS subquery to check if the same patient was readmitted within 30 days of their discharge. Then we count total admissions per insurance and how many of those had a qualifying readmission. Finally, we calculate the 30-day readmission percentage per insurance.
- Medicare and Medicaid hold the highest 30-day readmission rates.

Query 10 – Results

Row	insurance	total_admissions	readmit_30d	pct_readmit_30d
1	Medicare	28215	1993	7.06
2	Medicaid	5785	397	6.86
3	Private	22582	1018	4.51
4	Government	1783	70	3.93
5	Self Pay	611	22	3.6

Query 10 -

```sql
SELECT
  a.insurance,
  COUNT(*) AS total_admissions,
  SUM(
    CASE
      WHEN EXISTS (
        SELECT 1
        FROM `physionet-data.mimiciii_clinical.admissions` b
        WHERE b.subject_id   = a.subject_id
          AND b.admittime    > a.dischtime
          AND TIMESTAMP_DIFF(b.admittime, a.dischtime, DAY) <= 30
      )
      THEN 1 ELSE 0
    END
  ) AS readmit_30d,
  ROUND(
    100 * SUM(
      CASE
        WHEN EXISTS (
          SELECT 1
          FROM `physionet-data.mimiciii_clinical.admissions` b
          WHERE b.subject_id   = a.subject_id
            AND b.admittime    > a.dischtime
            AND TIMESTAMP_DIFF(b.admittime, a.dischtime, DAY) <= 30
        ) THEN 1 ELSE 0
      END
    )
    / COUNT(*)
  , 2) AS pct_readmit_30d
FROM
  `physionet-data.mimiciii_clinical.admissions` a
GROUP BY
  a.insurance
ORDER BY
  pct_readmit_30d DESC;
```

Broader Theme –

- Medicare and private insurance cover roughly 90% of ICU admissions, and most patients arrive through the emergency department.
- Medicare patients have the highest average number of ICU stays per admission, while Medicaid patients have the longest ICU length of stay.
- Medicare and private-insured patients have the most procedures on average, and Medicaid patients have the longest overall hospital stays.
- In-hospital mortality and 30-day readmission rates peak for Medicare and Medicaid, so there are payer-driven disparities in patient outcomes.